

Improved Automated Model Generation and Agent-Group specific Model Analysis

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Improved Automated Model Creation

The creation of rich multi-modal MATSim models is a challenging process with many things that could go wrong, unintentionally and sometimes even unknowingly.

Since its introduction at the MATSim User Meeting 2025, the scenario-generation service Creario got enhanced in various aspects.

Modelling of external commuter traffic. Every model has spatial boundaries, but in many places traffic does not stop at those boundaries. Creario is now able to create travel demand from outside the selected area, and include agents that travel inside the study area, for more complete travel demand. Agents from outside the study area receive their own subpopulations so they can be separately optimized, calibrated and analyzed.

Turn restrictions. For nearly two decades, MATSim intersections were very simple where vehicles could essentially turn in any direction as long as an out-link existed. Since 2023, MATSim supports turn restrictions, and Creario is now able to extract the necessary data from OpenStreetMap and include them in the generated MATSim networks, resulting in higher realism of the generated models.

Faster Model Creation. Various improvements in the automated model generation pipeline now allow us to create smaller models typically within minutes to one hour, while larger models only take a few hours. Being able to create even large, complex, multimodal scenarios within a single day is a game-changer and opens up the use of agent-based models to new category of projects.

The model creation pipeline is actually so efficient that **smaller models will be created for free** and fully automated.

Advanced Agent-Group specific Model Analysis

One big advantage of agent-based transport models over four-step models is the disaggregated travel demand. This allows to build arbitrary aggregations at the time of analysis. Tramola, the modelling suite for MATSim, focused on providing advanced and easy-to-use editing capabilities and integrated and scalable run execution in the past. Since the last user meeting, analysis and visualization got a larger refresh in Tramola with advanced visualizations even for large simulations and powerful analyses. The new analyses especially allow to filter results by agent attributes, making the promise of agent-based models finally available in an easy-to-use interface.